

where $\psi(x) = e^{|x|}(1 + |x|) - 1$.

We construct the coupling recursively. For the base case with $t = 0$, clearly $\mathbb{P}\{G_e^0 = T_e^0\} = 1$.

Condition on $T_e^{2t} = G_e^{2t}$ (with an appropriate vertex mapping) and event C^{2t} . We aim to construct a coupling so that $T_e^{2t+1} = G_e^{2t+1}$ and $T_e^{2t+2} = G_e^{2t+2}$ with probability at least $1 - n^{-\Omega(1)}$.

Each vertex u in ∂G_e^{2t} has B_u number of incident blue edges connecting to vertices in $[n] \setminus V(G_e^{2t})$, where the B_u 's are i.i.d. $\text{Binom}(n - |V(G_e^{2t})|, \lambda/n)$. Similarly, each vertex u in ∂T_e^{2t} has $\tilde{B}_u$ number of incident blue edges, where the $\tilde{B}_u$'s are i.i.d. $\text{Pois}(\lambda)$. Thus, we can couple B_u 's to $\tilde{B}_u$'s using (26) and take a union bound over $u \in \partial G_e^{2t} \equiv \partial T_e^{2t}$. In particular,

$$\begin{aligned} & \mathbb{P}\left\{B_u = \tilde{B}_u, \forall u \in \partial G_e^{2t} \mid G_e^{2t} = T_e^{2t}, C^{2t}\right\} \\ & \geq 1 - |\partial G_e^{2t}| \left(\lambda^2/n + \psi(\lambda - (n - |V(G_e^{2t})|)\lambda/n)\right) \\ & \geq 1 - (2k\lambda + 2)^t \log n \left(\lambda^2/n + O((2k\lambda + 1)^{t+1}\lambda/n)\right) \\ & \geq 1 - n^{-1+o(1)}, \end{aligned}$$

where the second inequality holds because conditional on C^{2t} , $|\partial G_e^{2t}| \leq (2k\lambda + 2)^t \log n$ and $|V(G_e^{2t})| \leq (2k\lambda + 1)^{t+1} \log n$. Thus, we have constructed a coupling such that $B_u = \tilde{B}_u$ for all $u \in \partial G_e^{2t}$ with probability at least $1 - n^{-1+o(1)}$.

Recall that if event E^{2t} occurs, the set of blue edges added to G_e^{2t+1} connect to distinct vertices in $[n] \setminus V(G_e^{2t})$. Thus, on event $E^{2t} \cap \{B_u = \tilde{B}_u, \forall u \in \partial G_e^{2t}\}$, there exists a one-to-one mapping from the vertices in ∂G_e^{2t+1} to vertices in ∂T_e^{2t+1} such that $G_e^{2t+1} = T_e^{2t+1}$. Further, recall that on event E^{2t+1} , each vertex u in ∂G_e^{2t+1} has exactly k incident red edges, and these red edges connect to distinct vertices in $[n] \setminus V(G_e^{2t+1})$. Thus, on the event $E^{2t+1} \cap E^{2t} \cap \{B_u = \tilde{B}_u, \forall u \in \partial G_e^{2t}\}$, there exists a one-to-one mapping from the vertices in ∂G_e^{2t+2} to the vertices in ∂T_e^{2t+2} , so that $G_e^{2t+2} = T_e^{2t+2}$. In conclusion, we get that

$$\begin{aligned} & \mathbb{P}\left\{G_e^{2t+2} = T_e^{2t+2} \mid G_e^{2t} = T_e^{2t}, C^{2t}\right\} \\ & \geq \mathbb{P}\left\{E^{2t+1} \cap E^{2t} \cap \{B_u = \tilde{B}_u, \forall u \in \partial G_e^{2t}\} \mid G_e^{2t} = T_e^{2t}, C^{2t}\right\} \\ & \geq \mathbb{P}\left\{B_u = \tilde{B}_u, \forall u \in \partial G_e^{2t} \mid G_e^{2t} = T_e^{2t}, C^{2t}\right\} - \mathbb{P}\left\{(E^{2t+1} \cap E^{2t})^c \mid G_e^{2t} = T_e^{2t}, C^{2t}\right\} \\ & \geq 1 - n^{-1+o(1)}, \end{aligned}$$

where the last inequality holds by Lemma F.3, since we are assuming $(2k\lambda + 2)^t \log n = n^{o(1)}$. Moreover,

$$\begin{aligned} & \mathbb{P}\left\{G_e^{2t+2} = T_e^{2t+2}, C^{2t+2} \mid G_e^{2t} = T_e^{2t}, C^{2t}\right\} \\ & \geq \mathbb{P}\left\{G_e^{2t+2} = T_e^{2t+2} \mid G_e^{2t} = T_e^{2t}, C^{2t}\right\} - (1 - \mathbb{P}\{C^{2t+2} \mid G_e^{2t} = T_e^{2t}, C^{2t}\}) \\ & \geq 1 - n^{-\Omega(1)}, \end{aligned}$$

where the last inequality holds by combining the last displayed equation with Lemma F.2. It